Question 1

Task 1

Entity	Description
Department	Represents an academic department in the university that offers courses and employs instructors
Course	Represents a subject offered by a department. Each course has details such as title, credit hours, and course level.
Instructor	Represents a faculty member responsible for teaching course sections and belonging to a department
Student	Represents a student enrolled in the university and associated with a department
Course Offering	Represents a specific offering of a course in a particular semester, year, and section, taught by an instructor.
Registration	Represents the enrollment of a student in a course section. It resolves the many-to-many relationship between Student and Course Offering and stores registration details such as registration date, enrollment status, and final grade.
Prerequisite	Represents the prerequisite relationship between courses, indicating that completion of one course is required before registering for another course.

Task 2

Entity	Attributes
Department	DepartmentID (PK), DepartmentName, OfficeLocation, ContactNumber
Course	CourseID (PK), CourseTitle, CreditHours, CourseLevel, DepartmentID (FK)
Instructor	InstructorID (PK), InstructorName, EmailAddress, PhoneNumber, Designation, Salary, DepartmentID (FK)
Student	StudentID (PK), StudentName, DateOfBirth, EmailAddress, PhoneNumber, AdmissionDate, AcademicStatus, DepartmentID (FK)
CourseOffering (Course Section)	OfferingID (PK), CourseID (FK), Semester, AcademicYear, SectionNumber, Classroom, Schedule, InstructorID (FK)
Registration	RegistrationID (PK), StudentID (FK), OfferingID (FK), RegistrationDate, EnrollmentStatus, FinalGrade
Prerequisite	CourseID (PK, FK), PrerequisiteCourseID (PK, FK)

Task 3

Entity	Primary Key (PK)	Why it Uniquely Identifies Each Record
Department	DepartmentID	Each department is assigned a unique DepartmentID, ensuring that no two departments can have the same identifier.
Course	CourseID	Each course has a unique CourseID that distinguishes it from all other courses.
Instructor	InstructorID	Each instructor is assigned a unique InstructorID, allowing the university to identify every instructor individually.
Student	StudentID	Each student receives a unique StudentID that uniquely identifies them within the university system.
CourseOffering (Course Section)	OfferingID	Each course offering or section has a unique OfferingID, allowing multiple sections of the same course to be distinguished.
Registration	RegistrationID	Each registration record is assigned a unique RegistrationID to identify a student's enrollment in a specific course section.

Prerequisite	(CourseID, PrerequisiteCourseID)	A composite primary key is used because the combination of a course and its prerequisite must be unique. This prevents the same prerequisite relationship from being entered more than once.

Task 4

Foreign Key Attribute	Referenced Table	Referenced Primary Key
DepartmentID (in Course)	Department	DepartmentID
DepartmentID (in Instructor)	Department	DepartmentID
DepartmentID (in Student)	Department	DepartmentID
CourseID (in CourseOffering)	Course	CourseID
InstructorID (in CourseOffering)	Instructor	InstructorID
StudentID (in Registration)	Student	StudentID
OfferingID (in Registration)	CourseOffering	OfferingID
CourseID (in Prerequisite)	Course	CourseID
PrerequisiteCourseID (in Prerequisite)	Course	CourseID

Task 5

Entity	Attribute	Data Type	Length/Range	NULL Allowed?
Department	DepartmentID	INT	Positive Integer	No
Department	DepartmentName	VARCHAR	100	No
Department	OfficeLocation	VARCHAR	100	No
Department	ContactNumber	VARCHAR	15	No
Course	CourseID	INT	Positive Integer	No
Course	CourseTitle	VARCHAR	100	No
Course	CreditHours	INT	1-6	No
Course	CourseLevel	VARCHAR	20	No
Course	DepartmentID	INT	Positive Integer	No
Instructor	InstructorID	INT	Positive Integer	No

Instructor	InstructorName	VARCHAR	100	No
Instructor	EmailAddress	VARCHAR	100	No
Instructor	PhoneNumber	VARCHAR	15	No
Instructor	Designation	VARCHAR	50	No
Instructor	Salary	DECIMAL(10,2)	≥ 0	No
Instructor	DepartmentID	INT	Positive Integer	No
Student	StudentID	INT	Positive Integer	No
Student	StudentName	VARCHAR	100	No
Student	DateOfBirth	DATE	Valid Date	No
Student	EmailAddress	VARCHAR	100	No
Student	PhoneNumber	VARCHAR	15	No
Student	AdmissionDate	DATE	Valid Date	No
Student	AcademicStatus	VARCHAR	20	No
Student	DepartmentID	INT	Positive Integer	No
CourseOffering	OfferingID	INT	Positive Integer	No
CourseOffering	CourseID	INT	Positive Integer	No
CourseOffering	Semester	VARCHAR	20	No
CourseOffering	AcademicYear	YEAR/INT	2000-2100	No
CourseOffering	SectionNumber	VARCHAR	10	No
CourseOffering	Classroom	VARCHAR	20	No
CourseOffering	Schedule	VARCHAR	100	No
CourseOffering	InstructorID	INT	Positive Integer	No
Registration	RegistrationID	INT	Positive Integer	No
Registration	StudentID	INT	Positive Integer	No
Registration	OfferingID	INT	Positive Integer	No
Registration	RegistrationDate	DATE	Valid Date	No
Registration	EnrollmentStatus	VARCHAR	20	No
Registration	FinalGrade	VARCHAR	2	Yes
Prerequisite	CourseID	INT	Positive Integer	No
Prerequisite	PrerequisiteCourseID	INT	Positive Integer	No

Task 6

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DEPARTMENT(
   DepartmentID PK,
   DepartmentName,
   OfficeLocation,
   ContactNumber
)
COURSE(
   CourseID PK,
   CourseTitle,
   CreditHours,
   CourseLevel,
   DepartmentID FK
)
INSTRUCTOR(
   InstructorID PK,
   InstructorName,
   EmailAddress,
   PhoneNumber,
   Designation,
   Salary,
   DepartmentID FK
)
STUDENT(
   StudentID PK,
   StudentName,
   DateOfBirth,
   EmailAddress,
   PhoneNumber,
   AdmissionDate,
   AcademicStatus,
   DepartmentID FK
)
COURSEOFFERING(
   OfferingID PK,
   CourseID FK,
   Semester,
   AcademicYear,
   SectionNumber,
   Classroom,
   Schedule,
   InstructorID FK
)
REGISTRATION(
   RegistrationID PK,
   StudentID FK,
   OfferingID FK,
   RegistrationDate,
```

```
   EnrollmentStatus,
   FinalGrade
)
PREREQUISITE(
   CourseID PK, FK,
   PrerequisiteCourseID PK, FK
)
```

Question 2:

I have attached sql file for this task